

est read range of the three types—passive, semi-active and active—these are the cheapest to manufacture and the easiest to integrate into objects. Passive tags typically operate in the 120-140kHz range. These are effective for building access control systems because they have a very short read range of 10 to 20 centimetres only.

Readers are complementary to tags and can be as technically diverse. The reader sends a pulse of energy to the tag and listens for the tag's response. The tag detects this energy and returns a response that contains the tag's serial number and possibly additional information. In simple RFID systems, the reader's energy pulse

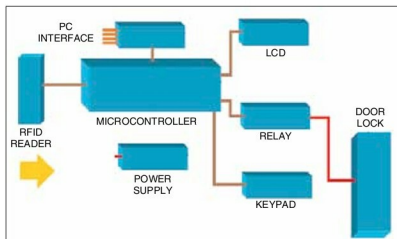


Fig. 6: System diagram of a microcontroller-based RFID building access control system

functions like an on-off switch. The size of the reader depends on many parameters and varies from the size of a coin to that of a PDA.

It is not very difficult to build your own access control system for your home or office building. You

can fabricate a simple microcontroller-based RFID building access control system using a pre-built RFID reader unit or reader module to read passive tags. When a tag is placed near the reader, the reader reads the tag and transfers the unique tag ID to the microcontroller. The microcontroller processes the data by comparing the received tag ID with

the saved ID. If the IDs match, the ID is displayed on the LCD panel in decimal number and the doorlock released by the system. ●

The author is a freelance writer and regular contributor to EFY

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